

• LIVE+RECORDED • DEEP DIVE

Maths for ML — Deep Dive 8 weeks

Calculus, Linear Algebra, Probability & Statistics — thoroughly

PRICE	DURATION	LEVEL	FORMAT	SCHEDULE
₹12,999	8 weeks	Intermediate	Live+Recorded	Ongoing · Sat & Sun · 10:00 AM – 12:00 PM IST, New batches start monthly



Prasant Mishra

Founder & Instructor, Qurious Academy

ABOUT THIS COURSE

What is Maths for ML — Deep Dive?

Eight weeks of rigorous mathematical foundations for machine learning. Goes significantly deeper than the weekend cohort — with proofs, derivations, and applied exercises every week.

IS THIS RIGHT FOR YOU?

Who Should Enroll

✓ THIS IS FOR YOU IF...

- ✓ You're someone with foundational knowledge ready to go further in Mathematics
- ✓ You want to learn by doing — with projects, exercises, and live instruction rather than passive video-watching
- ✓ You value efficiency: you'd rather spend focused hours on the right material than months on self-directed study
- ✓ You want expert feedback and a structured environment to stay accountable

✗ THIS IS NOT FOR YOU IF...

- ✗ You don't yet meet the prerequisites: 12th grade maths
- ✗ You prefer entirely self-paced learning with no live component — this course includes scheduled live sessions
- ✗ You already have advanced expertise in this area and are looking for highly specialised material

Prerequisites: 12th grade maths. Maths for ML Weekend Cohort or equivalent.

4 Sessions — 8 weeks

WEEKS 1-2 -
WEEKS 3-4

Calculus for ML & More

- Limits, continuity, differentiability
- Partial derivatives & the gradient
- Chain rule & backpropagation derivation
- Multivariable optimisation & Lagrange multipliers
- Vector spaces, basis & dimension
- Matrix factorizations (LU, QR, SVD)
- Eigendecomposition in depth
- PCA derivation from scratch

WEEKS 5-6 -
WEEKS 7-8

Probability & Statistics & More

- Probability theory — sigma algebras, random variables
- Common distributions & their properties
- Bayesian inference & MLE/MAP
- Hypothesis testing & confidence intervals
- Entropy, KL divergence, mutual information
- These concepts in neural networks
- Reading ML paper math sections
- Capstone: derive a classic ML algorithm

LEARNING OUTCOMES

What You'll Walk Away With

01 Derive and understand core ML algorithms mathematically

02 Read ML papers comfortably

03 Apply maths directly in code (NumPy)

04 Have a solid foundation for PhD-level AI work

WHAT'S INCLUDED

Everything You Get

✓ 16 live sessions + all recordings

✓ Weekly problem sets with full solutions

✓ NumPy code notebooks

✓ Learner's Certificate

LOGISTICS

Schedule & Pricing

FORMAT

Live+Recorded

SCHEDULE

Ongoing · Sat & Sun · 10:00 AM – 12:00 PM IST, New batches start monthly

DURATION

8 weeks · 4 sessions

PRICE

₹12,999

CERT learner

YOUR INSTRUCTOR

About Prasant Mishra



Prasant Mishra

Founder & Instructor — Qurious Academy

GenAI Solutions Architect

15+ Yrs Teaching

2500+ Students

60+ Projects Delivered

Prasant approaches mathematics as a language for describing the world precisely — one that unlocks everything from machine learning to financial modelling. Maths for ML — Deep Dive is built on the principle that mathematical intuition comes before symbol manipulation, not after.

He's seen too many students memorise procedures without understanding why they work. Every concept in this course is introduced through motivation first: here's the problem, here's why the standard approach fails, here's why this idea solves it.

QuriousAcademy

Focused, live learning for curious minds. No fluff — just the concepts that matter.

Maths for ML — Deep Dive

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New batches start monthly · Rolling start dates

₹12,999

Enroll at quriousacademy.com →

or visit quriousacademy.com/courses

Questions? Reach us at hello@quriousacademy.com

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